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Vellore Institute of Technology

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## Final Assessment Test(FAT) - Apr/May 2025

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| Programme | **B.Tech.** | Semester | **Winter Semester 2024-25** |
| Course Code | **BCSE307L** | Faculty Name | **Prof. Anand M** |
| Course Title | **Compiler Design** | Slot | **B1+TB1** |
| | | Class Nbr | **CH2024250502764** |
| Time | **3 hours** | Max. Marks | **100** |

**Instructions To Candidates**

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

**Course Outcomes**

CO1: Apply the skills on devising, selecting, and using tools and techniques towards compiler design
CO2: Develop language specifications using context free grammars (CFG).
CO3: Apply the ideas, the techniques, and the knowledge acquired for the purpose of developing software systems.
CO4: Constructing symbol tables and generating intermediate code.
CO5: Obtain insights on compiler optimization and code generation.

**Answer all Questions (10 × 10 Marks)**

01. Construct a Deterministic Finite Automata (DFA) for the following regular expression

**(ab|c|a)(abc|c)*(bc)*(b|ac)(bca)** using direct method.

**Note**: using **nullable(), firstpos(), lastpos()** and **followpos()**. **[10 Marks]**

**[10] (CO1/K3)**

02. Using phases of compiler translate the given program construct: **[10 Marks]**

```
while (x <= 10)
  {
    total = total + x * 10;
    x++;
  }
```

**[10] (CO1/K2)**

03. Consider the following grammar with **N = {S, A, B, C, D}** and **T = { ; , id , = , +, - , *, / , ( , ) , const }** are the set of non-terminals and terminals respectively. Productions of the grammar is

**S → A ; S | A**
**A → id = B**
**B → B + C | B – C | C**
**C → C * D | C / D | D**
**D → (A) | id | const**

Assume that integer is the only data type.

(a) Obtain the FIRST and FOLLOW sets for each non-terminal. **[4 Marks]**
(b) Construct an LL(1) parsing table for the grammar. **[3 Marks]**
(c) Parse the string **id = const ; id = const * id [3 Marks]**

**[10] (CO2/K3)**

04. Check whether the following grammar is CLR or not.

**S→ 0S0 | 1S1 | 10**

Where N ={S } and T ={0, 1 } are the set of non-terminal and terminals respectively.

a) Generate canonical collection of LR (1) items. **[4 Marks]**
b) Construct the corresponding CLR parse table for the above grammar.**[4 Marks]**
c) Conclude the grammar is CLR or not with suitable justification. **[2 Marks]**

**[10] (CO2/K3)**

05. The productions and the corresponding semantic rules are given below. Identify the synthesized and inherited attributes.

$T \rightarrow BC$ { T.x := C.x , C.y = B.x }
$B \rightarrow int$ { B.x := int }
$B \rightarrow float$ { B.x := float}
$C \rightarrow [num]\ C_1$ { C.x := array(num.val, $C_1$.x )
$C_1$.y := C.y }
$C \rightarrow \varepsilon$ {$C_1$.x = C.y}

(a) Draw an annotated parse tree for the expression **int [3][4][5] [3 Marks]**
(b) Clearly indicate the synthesized and inherited attribute in each of the internal nodes (if present) of your annotated parse tree. **[2 Marks]**
(c) Show the order of evaluation (as appeared in the dependency graph) of this parse tree for obtaining the expression **int [3][4][5] [5 Marks]**

**[10] (CO3/K3)**

06. Write the Three Address Code (TAC) for the following C function: **[10 Marks]**

```
void Sort ( int a[ ], int n) {
    int i, j, t;
    for(i=1; i <=n-1: i++)
      {
        j = i;
        while (j > 0 && a[j] < a[j-1] ) {
            t = a[j];
            a[j] = a[j-1];
            a[j-1] = t;
          --j;
      }
  }
return;
}
```

**[10] (CO4/K4)**

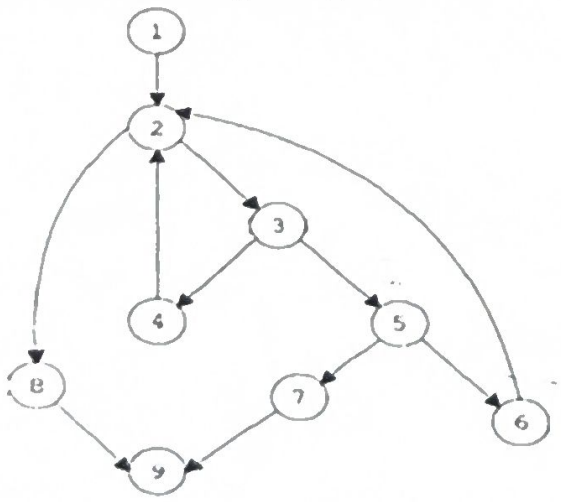
07. Derive a string **a + b * c + a * (c + b * a)** from the following grammar: **[2 Marks]**

$E \rightarrow E + T \mid T$
$T \rightarrow T * F \mid F$
$F \rightarrow ( E ) \mid a \mid b \mid c$

and translate the expression into
(a) Abstract syntax tree **[2 Marks]**
(b) Quadruple representation **[2 Marks]**
(c) Triple representation **[2 Marks]**
(d) Indirect triple representation **[2 Marks]**

**[10] (CO4/K5)**

08. From the following flow graph




(a) Identify the dominators **[4 Marks]**
(b) Identify the dominator tree **[4 Marks]**
(c) Identify the natural loops **[2 Marks]**

**[10] (CO5/K6)**

09. a) Construct the Directed Acyclic Graph (DAG) for the following Three Address Code: **[5 Marks]**

**u = u + v**
**v = v + w**
**x = v - w**
**y = v - x**
**z = u + v**

b) When the instructions are independent, is it possible to change the evaluation order during code generation phase? Justify your answer with an example. **[5 Marks]**

**[10] (CO5/K2)**

10. In the following machine code, R1, R2, R3 and R4 represent CPU registers. a and b represent memory locations. First operand is a destination and second operand is a source.

**LOAD R1, a**
**LOAD R2, b**
**MOV R3, R1**
**SUB R1, R2**
**MOV R4, R1**
**MOV R1, R3**
**ADD R1, R2**
**STORE a, R4**
**STORE b, R1**

a) For the given machine code, draw the data dependency graph. **[5 Marks]**

b) Assume that the MOV instruction consumes 1 clock cycle, ADD/SUB instructions consume 2 clock cycles and LOAD/STORE instructions consume 3 clock cycles. Consider the three machines 1, 2 and 3 with following instance of resources and find the shortest schedule for each machine to achieve maximum parallelism. **[5 Marks]**

Machine 1: one ALU resource and one MEM resource
Machine 2: one ALU resource and one MEM resources
Machine 3: two ALU resources and one MEM resource

**[10] (CO3/K3)**

**BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)**